

Reading a conjugate pair

A complex pair is not a failure of the real case — it is a rotation, and its two parts say by how much and through what angle. The modulus and argument are the whole content, and everything below is one of them applied.

RECOGNISING THE PAIR

3

1	When they appear 2×2 real matrices	$\text{tr}(A)^2 - 4 \det(A) < 0$ The discriminant settles it before any solving. A negative value means no real direction is mapped to a multiple of itself — the transformation turns every vector, so no real eigenvector can exist.
2	Conjugate pairing A real — not for complex entries	$\lambda = a + bi \Rightarrow \bar{\lambda} = a - bi$ Complex roots of a real polynomial arrive in pairs, and the eigenvectors pair with them: $\mathbf{v}$ and $\bar{\mathbf{v}}$. Nothing has to be computed twice — finding one half of the pair gives the other by conjugation.
3	Odd dimension real matrices only	n odd $\Rightarrow$ at least one real λ Complex roots come in pairs, so an odd number of them cannot pair off entirely and at least one must be real. This is why a rotation in $\mathbb{R}^3$ has an axis while a rotation in $\mathbb{R}^2$ has none — the leftover real eigenvalue is the axis.

READING THE PAIR

3

4	Modulus the scaling factor	$r = \sqrt{a^2 + b^2}$ How much the transformation stretches the plane per application. Since $r^2 = \det$ of the 2×2 block, the determinant is the area scaling and the modulus is its square root — consistent with the geometric reading .
5	Argument the rotation angle	$\theta = \arctan(b/a)$ How far the plane turns per application. A pair with $a = 0$ is a quarter turn; the sign of b picks the direction, and the two conjugates give the two orientations of the same rotation.
6	Real canonical form similar over $\mathbb{R}$, no complex arithmetic	$\begin{bmatrix} a & b \\ -b & a \end{bmatrix} = r \cdot R(\theta)$ The pair does not have to be handled in $\mathbb{C}$ at all. Over $\mathbb{R}$ the matrix is similar to a scaled rotation block, so the whole analysis stays real — which is what makes complex eigenvalues usable in applied work.

WHAT THE PAIR PREDICTS

2

7	Discrete stability for $\mathbf{x}_{k+1} = A\mathbf{x}_k$	$ \lambda < 1$ for every eigenvalue The modulus alone decides whether iteration spirals inward or outward; the argument only sets how fast it turns. Note $ \lambda = 1$ is the boundary case — a pure rotation that neither grows nor decays.
8	Continuous stability for $\dot{\mathbf{x}} = A\mathbf{x}$	$\text{Re}(\lambda) < 0$ for every eigenvalue A different criterion for a different system, and confusing the two is the usual error. Discrete systems compare the modulus against one; continuous systems compare the real part against zero — because the solution is $e^{\lambda t}$ and its growth is governed by $\text{Re}(\lambda)$.

The single idea underneath: a real matrix with eigenvalues $re^{\pm i\theta}$ acts on the corresponding real plane as a rotation by θ scaled by r . Recognition, canonical form and stability are all that statement read at different moments — which is why the modulus and argument are worth computing before anything else.